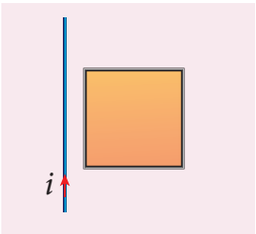
EXAMPLE 4.6

If the current i flowing in the straight conducting wire as shown in the figure decreases, find out the direction of induced current in the metallic square loop placed near it.

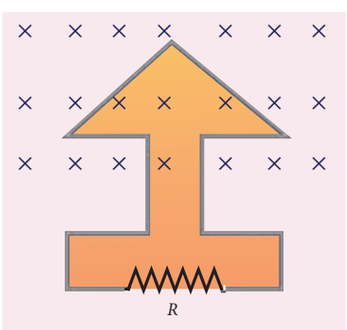


Solution

From right hand rule, the magnetic field by the straight wire is directed into the plane of the square loop perpendicularly and its magnetic flux is decreasing. The decrease in flux is opposed by the current induced in the loop by producing a magnetic field in the same direction as the magnetic field of the wire. Again from right hand rule, for this inward magnetic field, the direction of the induced current in the loop is clockwise.

EXAMPLE 4.7

The magnetic flux passes perpendicular to the plane of the circuit and is directed into the paper. If the magnetic flux varies with respect to time as per the following relation: $\Phi_B = \left(2t^3 + 3t^2 + 8t + 5\right) mWb$, what is the magnitude of the induced emf in the loop when $t = 3$ s? Find out the direction of current through the circuit.



Solution

$\Phi_B = \left(2t^3 + 3t^2 + 8t + 5\right) mWb$; $N = 1$; $t = 3$ s

(i) $$\varepsilon = \frac{d\left(N\Phi_B\right)}{dt}$$

$$= \frac{d}{dt}\left(2t^3 + 3t^2 + 8t + 5\right) \times 10^{-3}$$

$$= \left(6t^2 + 6t + 8\right) \times 10^{-3}$$

At t = 3 s,

$$\varepsilon = \left[\left(6 \times 9\right) + \left(6 \times 3\right) + 8\right] \times 10^{-3}$$

$$= 80 \times 10^{-3} V \quad = 80 mV$$

(ii) As time passes, the magnetic flux linked with the loop increases. According to Lenz's law, the direction of the induced current should be in a way so as to oppose the flux increase. So, the induced current flows in such a way to produce a magnetic field opposite to the given field. This magnetic field is perpendicularly outwards. Therefore, the induced current flows in anti-clockwise direction.

4.1.6 Motional emf from Lorentz force

Consider a straight conducting rod AB of length l in a uniform magnetic field $\vec{B}$ which is directed perpendicularly into the plane of the paper as shown in Figure 4.9(a). The length of the rod is normal to the magnetic field. Let the rod move with a constant velocity $\vec{v}$ towards right side.

When the rod moves, the free electrons present in it also move with same

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